

# HT-Advisor technician heat-treatment sheet

Fatigue objective - LPBF Inconel 718 - non-HIP route

## Input conditions

| Field                        | Value                          |
|------------------------------|--------------------------------|
| Primary objective            | fatigue                        |
| Alloy and process            | LPBF Inconel 718               |
| Initial material state       | EOS-like LPBF, as-built        |
| Build orientation            | vertical                       |
| Surface condition            | machined                       |
| Representative section size  | thin coupon                    |
| HIP benchmark included       | No                             |
| Maximum furnace temperature  | 1100 C                         |
| Maximum practical cycle time | 20 h                           |
| Fatigue validation target    | R = 0.1; Nf = 1,000,000 cycles |

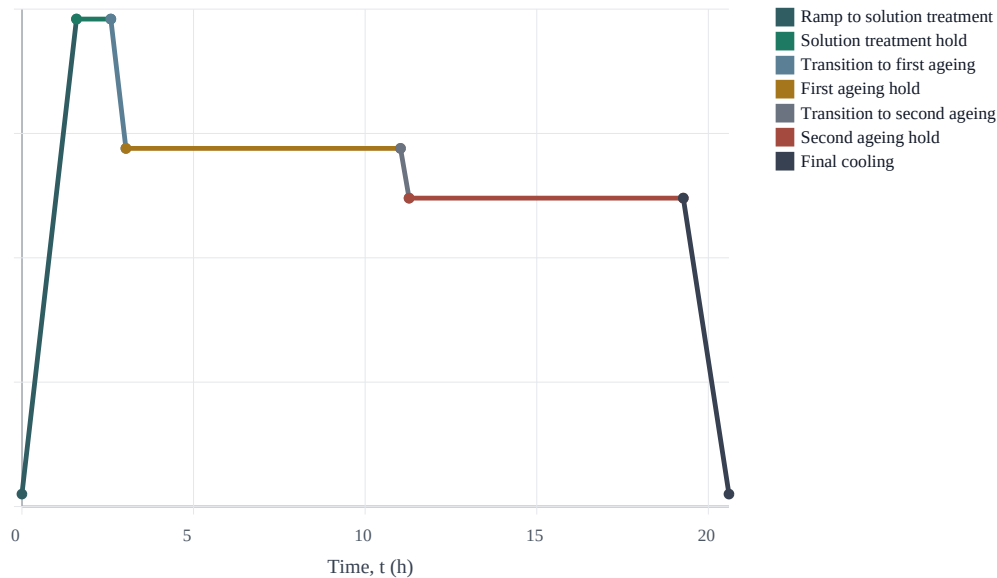
## Recommended route

| Field                     | Value                                       |
|---------------------------|---------------------------------------------|
| Route                     | ST_DA                                       |
| Approved recipe           | 980 C for 1 h; 720 C for 8 h; 620 C for 8 h |
| Peak temperature          | 980 C                                       |
| Total specified hold time | 17.0 h                                      |
| Recommendation index      | 0.77                                        |
| Local feasibility         | feasible under selected constraints         |

## Nominal thermal programme

| Step | Set point | Hold time | Action              |
|------|-----------|-----------|---------------------|
| 1    | 980 C     | 1.0 h     | Hold at temperature |
| 2    | 720 C     | 8.0 h     | Hold at temperature |
| 3    | 620 C     | 8.0 h     | Hold at temperature |

### Thermal step profile: temperature T (C)



The thermal profile is nominal. Record actual ramp rate, soak start and end times, cooling condition, specimen placement, and thermocouple or witness-coupon position.

### Fatigue validation schedule

| sigma_a (MPa) | R   | sigma_max (MPa) | sigma_min (MPa) | Target runout |
|---------------|-----|-----------------|-----------------|---------------|
| 300           | 0.1 | 667             | 67              | 1000000       |
| 350           | 0.1 | 778             | 78              | 1000000       |
| 400           | 0.1 | 889             | 89              | 1000000       |
| 450           | 0.1 | 1000            | 100             | 1000000       |

### Required process records

- Furnace ID: to be completed.
- Furnace programme ID: to be completed.
- Furnace atmosphere, vacuum, or shielding gas: to be completed.
- Thermocouple or witness coupon location: to be completed.
- Final cooling method: controlled furnace cooling unless locally approved otherwise.
- Operator sign-off and process-owner approval: to be completed.

### Must-have experimental validation

- required: as-built baseline - Quantifies the starting LPBF Inconel 718 response before any thermal treatment.
- required: AMS-style standard baseline - Compares the framework-recommended route against a conventional solution plus double-ageing reference route.
- required: framework-recommended route: ST\_DA - Tests the route selected by the decision-support workflow under the local furnace constraints.
- required: hardness and tensile testing - Validates the static property indicators used by the current metallurgy-informed model.
- required: SEM/EDS microstructural assessment - Checks Laves/Nb-rich segregation, precipitate response, and evidence that the selected heat treatment produced the intended metallurgical change.
- strongly recommended: fatigue or staircase screening if machine time is available - Non-HIP LPBF fatigue is defect-controlled; static tensile indicators alone do not establish fatigue safety.

## **S-N model boundary**

Fatigue life is not predicted in the current release; stress ratio and target cycles are used only to plan validation tests.